Justifying constructions using congruent triangles

1 Angle bisector

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a $\overline{AC}, \overline{BC}$

b $\overline{AB}, \overline{CB}$

c Equilateral

3

a The arcs that created the segments have the same radius.

b Yes, by the side-side-side congruence theorem.

c

i Yes

ii No

iii No

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To prove: $\angle EDF \cong \angle BAC$

	Statements	Reasons
1.	$\overline{AB} \cong \overline{AC} \cong \overline{DE} \cong \overline{DF}$	The arcs that created the segments have the same radius.
2.	$\overline{BC} \cong \overline{EF}$	The arcs that created the segments have the same radius.
3.	$\triangle ABC \cong \triangle DEF$	Side-side-side congruence of triangles
4.	$\angle BAC \cong \angle EDF$	Corresponding parts of congruent triangles are congruent (CPCTC).

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To prove: $\overleftrightarrow{AC} \parallel \overleftrightarrow{DF}$

	Statements	Reasons
1.	$\overline{AC} \cong \overline{BC} \cong \overline{DF} \cong \overline{EF}$	The arcs that created the segments have the same radius.
2.	$\overline{AB} \cong \overline{DE}$	The arcs that created the segments have the same radius.
3.	$\triangle ABC \cong \triangle DEF$	Side-side-side congruence of triangles
4.	$\triangle ACB \cong \triangle DFE$	Corresponding parts of congruent triangles are congruent (CPCTC).
5.	$\overleftrightarrow{AC} \parallel \overleftrightarrow{DF}$	Converse of corresponding angles theorem

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To prove: $\overrightarrow{AD}$ is an angle bisector

	Statements	Reasons
1.	$\overline{AB} \cong \overline{AC} \cong \overline{BD} \cong \overline{CD}$	The arcs that created the segments have the same radius.
2.	$\overline{AD} \cong \overline{AD}$	Reflexive property of congruence
3.	$\triangle ABD \cong \triangle ACD$	Side-side-side congruence of triangles
4.	$\angle BAD \cong \angle CAD$	Corresponding parts of congruent triangles are congruent (CPCTC).
5.	$\overrightarrow{AD}$ is an angle bisector	Definition of an angle bisector

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To prove: $\overrightarrow{DB} \parallel \overleftrightarrow{AC}$

	Statements	Reasons
1.	$\overline{AD} \cong \overline{CD}$	The arcs that created the segments have the same radius.
2.	$\overline{AB} \cong \overline{CB}$	The arcs that created the segments have the same radius.
3.	$\overline{BD} \cong \overline{BD}$	Reflexive property of congruence
4.	$\triangle ABD \cong \triangle CBD$	Side-side-side congruence of triangles
5.	$\triangle ADB \cong \triangle CDB$	Corresponding parts of congruent triangles are congruent (CPCTC).
6.	$m\angle ADB = m\angle CDB = 90^\circ$	Angles that are both a linear pair and congruent must be right angles.
7.	$\overrightarrow{DB} \parallel \overleftrightarrow{AC}$	Definition of perpendicular

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To prove: $\overleftrightarrow{BD} \perp \overleftrightarrow{AC}$

	Statements	Reasons
1.	$\overline{AB} \cong \overline{CB}$	The arcs that created the segments have the same radius.
2.	$\overline{AD} \cong \overline{CD}$	The arcs that created the segments have the same radius.
3.	$\overline{BD} \cong \overline{BD}$	Reflexive property of congruence
4.	$\triangle ABD \cong \triangle CBD$	Side-side-side congruence of triangles
5.	$\angle ABE \cong \angle CBE$	Corresponding parts of congruent triangles are congruent (CPCTC).
6.	$\overline{BE} \cong \overline{BE}$	Reflexive property of congruence
7.	$\triangle ABE \cong \triangle CBE$	Side-angle-side congruence of triangles
8.	$\angle AEB \cong \angle CEB$	Corresponding parts of congruent triangles are congruent (CPCTC).
9.	$m\angle AEB = m\angle CEB = 90^\circ$	Angles that are both a linear pair and congruent must be right angles.
10.	$\overleftrightarrow{BD} \perp \overleftrightarrow{AC}$	Definition of perpendicular

9 a

To prove: E [redacted] **midpoint of** $\overline{AB}$

	Statements	Reasons
1.	$\overline{AC} \cong \overline{AD} \cong \overline{BC} \cong \overline{BD}$	The arcs that created the segments have the same radius.
2.	$\overline{CD} \cong \overline{CD}$	Reflexive property of congruence
3.	$\triangle ACD \cong \triangle BCD$	Side-side-side congruence of triangles
4.	$\angle ACE \cong \angle BCE$	Corresponding parts of congruent triangles are congruent (CPCTC).
5.	$\overline{CE} \cong \overline{CE}$	Reflexive property of congruence
6.	$\triangle ACE \cong \triangle BCE$	Side-angle-side congruence of triangles
7.	$\overline{AE} \cong \overline{BE}$	Corresponding parts of congruent triangles are congruent (CPCTC).
8.	E is the midpoint of $\overline{AB}$	Definition of midpoint

b

To prove: $\overleftrightarrow{CD}$ bisects $\overline{AB}$ and $\overleftrightarrow{AB}$ bisects $\overline{CD}$

	Statements	Reasons
1.	$\overline{AC} \cong \overline{AD} \cong \overline{BC} \cong \overline{BD}$	The arcs that created the segments have the same radius.
2.	$\overline{AB} \cong \overline{AB}$	Reflexive property of congruence
3.	$\triangle ACB \cong \triangle ADB$	Side-side-side congruence of triangles
4.	$\overline{CD} \cong \overline{CD}$	Reflexive property of congruence
5.	$\triangle ACD \cong \triangle BCD$	Side-side-side congruence of triangles
6.	$\angle ACE \cong \angle BCE, \angle CBE \cong \angle DBE, \angle BDE \cong \angle ADE$ and $\angle DAE \cong \angle CAE$	Corresponding parts of congruent triangles are congruent (CPCTC).
7.	$\triangle ACE \cong \triangle BCE \cong \triangle BDE \cong \triangle ADE$	Side-angle-side congruence of triangles
8.	$\angle AEC \cong \angle BEC \cong \angle BED \cong \angle AED$	Corresponding parts of congruent triangles are congruent (CPCTC).
9.	$m\angle AEC \cong m\angle BEC \cong m\angle BED \cong m\angle AED = 90$	Angles that are both a linear pair and congruent must be right angles.
10.	$\overleftrightarrow{CD} \perp \overleftrightarrow{AB}$	Definition of perpendicular
11.	$\overline{AE} \cong \overline{BE} \cong \overline{CE} \cong \overline{DE}$	Corresponding parts of congruent triangles are congruent (CPCTC).
12.	$\overleftrightarrow{CD}$ bisects $\overline{AB}$ and $\overleftrightarrow{AB}$ bisects $\overline{CD}$	Definition of bisector